

GRAPHIC ERA (DEEMED TO BE UNIVERSITY), DEHRADUN

SEMESTER I

Name of Department: - Department of Computer Applications

Course Code:		TMC104	Course Title:		Advance Operating Systems		
Credits:	3	Semester:	I	Category of Course:		DSC	
TEACHING PERIODS (In hours per week)			TW+SL (in hours per semester)		Relative Weight		
L	T	P	45		CIE	MSE	ESE
3	0	0			25	25	50
Examination Duration (Hrs):			Theory			Practical	
			3			0	

Pre-requisite: To understand students with the Concept of Operating Systems that are used to manage various system components including files and processes.

Course Outcome:	After completion of the course the students will be able to: CO1: List the functions and purposes of modern operating systems. CO2: Describe the file and storage management concepts CO3: Differentiate between the protection and security features of an operating system and its implementation mechanisms CO4: To interpret the concept of virtualization and describe its advantages, and need. CO5: Evaluate the advantages, design issues, and robustness of distributed systems and parallel systems in operating environments. CO6: To analyze and compare the design, features, and functionality of Linux, and Windows operating systems through a case study
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Details of the Course:

Sl. No.	Contents	Contact Hours
1	Unit 1: Overview of OS Objectives and Functions, Kernel & Shell, Multiprogramming, & Multitasking Operating Systems, CPU Scheduling Algorithms (FCFS, Round Robin). SVR3 Scheduling Algorithms.	9

2	Unit 2: File and Storage Management: Files and File systems, File organization and Access, File Directory and Sharing, Secondary storage management. Protection and Security, Goals of Protection, Principles of Protection, Domain of Protection, Access Control.	9
3	Unit 3: Real-Time Operating System (RTOS) Background, Characteristics of Real-Time Operating Systems, Types of RTOS, RTOS kernel and function, Task Management, Real-Time Scheduling (Rate Monotonic, EDF), Application Example from Linux Operating System.	9
4	Unit 4: Distributed Systems: Advantages of Distributed Systems, Distributed Operating Systems, Types of Distributed Operating Systems, Robustness and Design Issues of Distributed OS, Parallel Systems Definition, Parallel vs Distributed Systems, Example, Types of Parallel Systems: Overview.	9
5	Unit 5: Virtualization: Definition, Advantages, and Needs, Types, Building Block Diagram, Examples, Cloud Operating Systems, Goals and working, Examples of Cloud Operating Systems. Case Studies/examples from UNIX/Linux/Windows operating systems.	9
	TOTAL	45

Text Books:

Authors Name	Title	Edition	Publisher, Country	Year
Abraham Silberschatz, Peter B Galvin, Greg Gagne	Operating System Concepts	9 th Edition	Wiley India Pvt. Ltd .	2018
William Stallings	Operating Systems Internals and Design Principles	9 th Edition	Pearson	2018

Reference Books:

Authors Name	Title	Edition	Publisher, Country	Year
Andrew S. Tanenbaum	Distributed Operating Systems	9 th Edition	Pearson	2018

Abraham Silberschatz, Peter B. Galvin & Greg Gagne	Operating System Concepts	10 th Edition	Wiley	2018
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